THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS TWO-CHAPTER TEST SET B

Chapters 5 & 6: **Number Play, We Distribute, Yet Things Multiply** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A number is divisible by 4 if the number formed by its last two digits is divisible by: 1

(a) 4 **(b)** 2 **(c)** 5 **(d)** 8

2. Which of these is divisible by 5? 1

(a) 3475 **(b)** 3471 **(c)** 3472 **(d)** 3476

3. Starting with 3 and subtracting the reverse of its digits repeatedly (Kaprekar), the process for 3-digit numbers ends at: 1

(a) 999 **(b)** 100 **(c)** 594 **(d)** 495

4. The HCF of 24 and 36 is: 1

(a) 6 **(b)** 72 **(c)** 24 **(d)** 12

5. The number 12345679 × 9 equals: 1

(a) 123456790 **(b)** 100000001 **(c)** 111111111 **(d)** 999999999

6. $(a-b)^2$ equals: 1

(a) $a^2 - ab + b^2$ **(b)** $a^2 - 2ab + b^2$ **(c)** $a^2 - b^2$ **(d)** $a^2 + 2ab + b^2$

7. $(3 \times 4) \times 5 = 3 \times (4 \times 5)$ illustrates the: 1

(a) distributive property **(b)** inverse property

(c) commutative property **(d)** associative property

8. The coefficient of x in $(x+5)(x+2)$ is: 1

(a) 5 **(b)** 3 **(c)** 7 **(d)** 10

9. The value of $25 \times 4 \times 7$ using the best grouping is: 1

(a) 1400 **(b)** 70 **(c)** 350 **(d)** 700

10. $(a-b)^2$ is always: 1

(a) zero **(b)** negative **(c)** odd **(d)** non-negative

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Find the digit x if $46x$ is divisible by 9. 2

12. Solve the cryptarithm: $A + A = 10$. Find A. 2

13. Simplify $15 \times 27 + 15 \times 73$. 2

14. Evaluate 52^2 using $(50+2)^2$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. A bell rings every 12 minutes and another every 18 minutes. If both ring together at 8 a.m., when will they next ring together? 3

16. Expand and simplify $(x+6)(x+8)$. Verify for $x=1$. 3

17. Show that the number formed by writing any digit three times, such as 777, is divisible by 3 and 37. 3

OR

The sum of the digits of a two-digit number is 9. Show that the number formed by reversing the digits differs by a multiple of 9. 3

18. Expand and simplify $(x+4)(x+11)$. Verify for $x=1$. 3

OR

Mind the mistake: a student writes $(x+5)^2 = x^2+25$. Find the mistake and correct it. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. **Case study:** A museum numbers tickets from 1001 to 1100. 5

(i) How many ticket numbers are divisible by 7?
(ii) Which are divisible by 11?
(iii) Which ticket is divisible by both?

OR

(i) Take 854, subtract its reverse, then add the reverse of the difference. 5
(ii) Repeat with 732 and state what you notice.

20. **Case study:** A student multiplies $(a+b)(a+b)$ as a^2+b^2. 5

(i) What is wrong?
(ii) Test with $a=4$, $b=1$.
(iii) Write the correct result.

OR

(i) Find the product $(x^2+2)(x^2-2)$. 5
(ii) Evaluate it at $x=3$.

END OF QUESTION PAPER — Total: 40 Marks

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